

Classifier-Free Diffusion Guidance for de-Novo AAV Capsid Design

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I. Introduction

Adeno-associated viruses (AAVs) have been identified as promising vehicles for gene therapy as they exhibit low pathogenicity and sustained cargo expression. They function by packaging a therapeutic DNA payload inside a protein capsid that binds to cell-surface receptors, enters the cell via endocytosis, escapes the endosome, and delivers its genome to the nucleus where the transgene is expressed [8]. Existing approaches for AAV capsid design have made progress but remain limited in scope. The Fit4Function framework, for example, learns sequence–function relationships over constrained peptide insertions (e.g., short surface motifs) to optimize traits such as manufacturability and tropism, but operates within a restricted local design space rather than enabling fully de novo capsid generation [5]. Generating AAVs with great structural diversity has also been cited as a major challenge [2]. To address the above, we propose optimization of AAV capsids via the development of 1) a predictor model capable of converting capsid protein sequence into scores for packaging viability and synthetic feasibility, and 2) a diffusion model with classifier-free generation capable of designing capsids with improvements on those same metrics.

II. Datasets

We will use the public AAV2 capsid dataset from Bryant et al., which studies a 28-amino-acid variable region of the VP3 capsid protein and measures packaging-related fitness/viability through viral selection assays. The published paper reports 201,426 generated variants including 110,689 viable engineered capsids with the associated data repository providing AAV packaging viability assay results (performed by researchers in vitro) and partitioned training/validation data.

In our preprocessing pipeline, we will standardize sequence fields across files, remove sequences with very low or zero plasmid counts, add pseudocounts to stabilize ratios, compute a log2 viral-selection score from virus/plasmid frequencies, derive a viability label from that score, and de-duplicate repeated designed sequences before combining the available chips into a unified training set. This preprocessing is moderate in scope rather than trivial, but it is already partly implemented in a python notebook included in paper’s code repository, so the main remaining effort is validating the merged schema and defining clean train/validation/test splits.

III. Methods

Numerous unconditional approaches have been proposed in the literature for high-throughput viable AAV capsid candidate generation [1, 2, 3]. In addition, various pipelines have been developed for new AAV capsid generation conditioned upon desirable traits. Eid et. al. developed a robust framework for predicting such traits and utilized this to score new candidates proposed by large-scale random sequence sampling [5]; this approach, however, constrains exploration to a predefined local sequence space. Xiao et al. proposes a framework in which diffusion denoising is guided by a viability classifier network [6]; however it is unclear whether this classifier is robust on noisy representations during diffusion.

For candidate generation, we will train a discrete diffusion model over the 28-AA variable region of AAV2 VP3 on the viable subset of Bryant et al., using classifier-free guidance (CFG) [4] as conditioning. During training, each sequence is paired with its measured fitness

score as a conditioning input, with the condition randomly dropped 15% of the time to encourage both conditional and unconditional denoising. At inference, we combine the two noise predictions as $\epsilon = \epsilon_{\text{uncond}} + w \cdot (\epsilon_{\text{cond}} - \epsilon_{\text{uncond}})$, where guidance scale $w > 1$ pushes samples toward higher fitness at the cost of diversity. We will sweep w to characterize this tradeoff. Unlike prior AAV diffusion work (AAVDiff, AAVDiffusion), which uses classifier-guidance-style gradients from a separate predictor, CFG bakes conditioning directly into the diffusion model and avoids the noisy-intermediate problem that affects classifier guidance. To evaluate proposed candidates generated by the CFG pipeline, we will develop a fitness score classifier trained on different data not seen by the diffusion model [7].

IV. Evaluation Criteria

We will evaluate both the predictive model and the generative model. For the predictor, we will measure performance on held-out sequences using classification metrics such as AUROC and AUPRC if we use a binary viability label, and correlation metrics such as Pearson or Spearman if we predict the continuous viral-selection score derived during preprocessing. Because the dataset is likely imbalanced between viable and non-viable variants, we will emphasize AUPRC in addition to AUROC. For the generative model, success will be measured by the fraction of generated sequences that are predicted to be viable or high-fitness by the held-out predictor, compared against baselines such as random sampling from the training distribution and unconditional generation. To ensure the model is not simply memorizing training examples, we will also evaluate diversity and novelty of generated sequences using metrics such as mean pairwise edit distance and minimum edit distance to the training set. Experimentally, we will use train/validation/test splits defined before model development and avoid evaluating generation with the same predictor used to guide training whenever possible. A feasible target is to build a predictor that generalizes well to held-out variants and distinguishes viable from non-viable sequences substantially better than chance. An ambitious target is to show that classifier-free-guided generation produces sequences with higher predicted viability than unconditional or random baselines while maintaining substantial sequence diversity. This would demonstrate that the model is learning useful structure in AAV capsid sequence space rather than only reproducing known viable examples. If we are unable to do so, a more modest target is simply the successful design of a model capable of predicting these traits given the sequence.

V. Analysis and Application

Within the biological molecule design domain, translating in-silico discovery to in-vitro results is often cost and time-prohibitive. While molecular dynamics simulations could narrow down candidates according to structural or mechanistic constraints, compute times often prohibitively scale with respect to the number of molecules. As a result, the burden for in-silico biomolecule discovery depends on more accurate and efficient means of candidate generation and assessment. Our proposed method for capsid sequence generation seeks to do this within the domain of gene therapy delivery vehicles, by leveraging classifier-free guidance, a modern method for diffusion-based candidate generation.

VI. References

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